

Studying Development of Psychopathology Across the Lifespan

Isaac T. Petersen, Zachary Demko, and Megan Schumer

Department of Psychological and Brain Sciences, University of Iowa

Author Note

Isaac T. Petersen  <https://orcid.org/0000-0003-3072-6673>

Zachary Demko  <https://orcid.org/0000-0002-5717-6205>

Megan Schumer  <https://orcid.org/0009-0009-5488-6541>

The data for the present study are published online: [LINK]. We have no conflicts of interest to disclose. The present study was approved by the University of Iowa Institutional Review Board (Study #: INSERT). The authors complied with APA ethical standards in the treatment of participants. I.P. was funded by Grant HD098235 from the Eunice Kennedy Shriver National Institute of Child Health and Human Development. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Institutes of Health. We thank Jordan L. Harris and Alexis Hosch for their help identifying measures of externalizing problems and transcribing and coding their content. Author roles were classified using the Contributor Role Taxonomy (CRediT; <https://credit.niso.org>) as follows: Isaac T. Petersen: conceptualization, data curation, analysis, funding, investigation, methodology, project administration, resources, software, supervision, validation, visualization, writing, editing; Zachary Demko: conceptualization, data curation, analysis, investigation, methodology, software, editing; Megan Schumer: editing

Correspondence concerning this article should be addressed to Isaac T. Petersen, University of Iowa, G60 Psychological and Brain Sciences Building, Iowa City, IA 52242. Email: isaac-t-petersen@uiowa.edu

Abstract

The field has not adequately studied individuals' development of psychopathology across the lifespan, instead taking a piecewise approach to development, limiting our understanding of lifespan development and the origins of psychopathology. We describe why this is the case—beyond logistical challenges. First, diagnostic and taxonomic systems do not define psychopathology constructs from a developmental perspective. A developmental perspective is crucial because psychopathology constructs often have earlier origins than their full clinical presentation, and forms of psychopathology frequently show changes in behavioral manifestation across development (heterotypic continuity). Second, current assessments do not adequately capture these changes in behavioral manifestation. Third, traditional scoring systems do not allow charting individuals' change over time in the face of changing behavioral manifestation. Fourth, assessments are not well-suited for capturing individual differences, despite the dimensional nature of psychopathology. We propose solutions to these challenges by introducing a developmental scaling framework for psychopathology. Using externalizing psychopathology as an example, we draw on our content analysis of externalizing measures, as well as lessons from the study of other constructs across lengthy developmental periods. Addressing these challenges will allow charting individuals' trajectories of psychopathology across the lifespan and identifying risk and protective factors that influence developmental courses.

Keywords: psychopathology, lifespan development, longitudinal trajectories, heterotypic continuity, externalizing behavior problems

Studying Development of Psychopathology Across the Lifespan

NOTE: this manuscript is in progress of being drafted—it is incomplete.

When entrenched, psychopathology is costly and difficult to treat (Insel, 2008; Wakschlag et al., 2019). Understanding its developmental origins is therefore crucial for preventing later, more severe psychopathology. Developmental science seeks to build a bridge that spans the lifespan—from infancy to older adulthood—and not just transitory periods in people's lives. However, the field has not adequately studied individuals' development of psychopathology across the lifespan, instead taking a piecewise approach to development that focuses on narrower age ranges. The focus on narrow age ranges restricts our ability to understand development across important transitions, such as puberty, school entry, and entry into parenthood, thus limiting our understanding of lifespan development and the origins of psychopathology. In this paper, we describe reasons why this is the case—beyond logistical challenges of time and money. And we propose solutions to these challenges by introducing a developmental scaling framework for psychopathology. Using externalizing psychopathology as an example, we draw on our content analysis of over 270 externalizing measures, as well as lessons from the study of other constructs such as academic and cognitive skills across lengthy developmental periods.

The Goal

A key goal of developmental science is to understand how people develop across the lifespan. A variety of methods, including cross-sectional and longitudinal designs, could be leveraged for this purpose. In a cross-sectional design, people are assessed at one timepoint. Cross-sectional designs provide the ability to cheaply and quickly assess a wide age range, providing a snapshot at a particular point in time. Cross-sectional designs may inform our understanding of general normative patterns of change—such as the typical pattern of cognitive growth and decline across the lifespan.

However, cross-sectional designs have key limitations. Cross-sectional designs do not allow examining individuals' change over time. Individuals show considerable heterogeneity in level and change across time in many constructs, so cross-sectional designs may obscure diverse

patterns of change. Moreover, in a cross-sectional study, any apparent age-related differences in a construct could reflect cohort differences instead of developmental changes. For example, average differences in stress between 5-year-olds and 10-year-olds could reflect the differing experiences that the 10-year-olds shared—such as going to school during the Coronavirus pandemic.

However, even cross-sectional work spanning the lifespan is not often conducted.

In a longitudinal design, the same individuals are followed across time.

The Problem/Obstacle

Reasons for the Problem

Beyond logistical challenges of time and money, there are several key reasons why psychology has failed to study people's development of psychopathology, longitudinally, across the lifespan. First, diagnostic and taxonomic systems do not define psychopathology constructs from a developmental perspective. A developmental perspective is crucial because psychopathology constructs often have earlier origins than their full clinical presentation, and forms of psychopathology frequently show changes in behavioral manifestation across development (heterotypic continuity). Second, current assessments do not adequately capture these changes in behavioral manifestation. Third, traditional scoring systems do not allow charting individuals' change over time in the face of changing behavioral manifestation. Fourth, current assessments are not well-suited for capturing the full range of individual differences, despite the dimensional nature of psychopathology. Fifth, as a result of these limitations, current assessments are not well-suited for capturing individuals' change over time.

Below, we describe these reasons in detail. The first reason represents problems with our constructs. The remaining reasons represent problems with existing assessments and traditional scoring systems.

Problems of our Constructs

Lack of a Developmental Perspective

Problems of Existing Assessments and Traditional Scoring Systems

Do Not Adequately Capture Constructs' Change in Behavioral Manifestation Across Development

Do Not Allow Charting Individuals' Change Over Time When the Construct Changes in Behavioral Manifestation

Are Not Well-Suited For Capturing the Full Range of Individual Differences

Are Not Well-Suited for Capturing Individuals' Change Over Time

Example: Externalizing Psychopathology

Solution: A Developmental Scaling Framework**Other Key Challenges to Address**

Potential Confounds of Change

Factorial Invariance Across Time

Obsolescence of Items and Measures

Conclusion

Addressing these challenges will allow charting individuals' trajectories of psychopathology across the lifespan and identifying risk and protective factors that influence their developmental courses. Doing so is critical for understanding how psychopathology develops, when to intervene, and how to most effectively prevent psychopathology.

References

- Insel, T. R. (2008). Assessing the economic costs of serious mental illness. *American Journal of Psychiatry*, 165(6), 663–665. <https://doi.org/10.1176/appi.ajp.2008.08030366>
- Wakschlag, L. S., Roberts, M. Y., Flynn, R. M., Smith, J. D., Krogh-Jespersen, S., Kaat, A. J., Gray, L., Walkup, J., Marino, B. S., Norton, E. S., & Davis, M. M. (2019). Future

directions for early childhood prevention of mental disorders: A road map to mental health, earlier. *Journal of Clinical Child & Adolescent Psychology*, 48(3), 539–554.

<https://doi.org/10.1080/15374416.2018.1561296>

Appendix A

This Section Is an Appendix

Appendix B

Another Appendix